

Mohammad Imrul Jubair

University of Colorado Boulder

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Education

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| 2022 – Present | PhD Student in Computer Science
University of Colorado Boulder, USA
Advisor: Prof. Tom Yeh |
| 2014 – 2016 | MSc in Computer Science
University of Calgary, Canada
Thesis: Icosahedral Maps for a Multiresolution Representation of Earth Data
Advisor: Prof. Usman Alim |
| 2008 – 2011 | BSc in Computer Science and Information Technology
Islamic University of Technology, Bangladesh |

Research Interests

Scientific visualization, explainable artificial intelligence, computer vision, and medical image analysis

Research Experience

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| 2022 – Present | Doctoral Researcher, Imagine AI Lab
University of Colorado Boulder
Current research: Developing interactive tools for understanding AI models, including byhand CLI, a terminal-based model visualization tool, and The Architect, an Excel-based system released as a 2026 technical report.
Previous research: Contributed to 3D cardiac MRI segmentation and the HVSMR-2.0 dataset, resulting in a 2024 publication in <i>Scientific Data</i> , a Nature Portfolio journal. |
| 2024 | Summer Research Assistant
University of Colorado Anschutz Medical Campus
Studied early detection of Alzheimer's and related dementias using financial activity data. |
| 2021 | Independent Research Collaborator
Investigated physics-based camera calibration methods in collaboration with Dr. Helge Rhodin of the University of British Columbia. |

2017 – 2021	Independent Researcher Conducted independent research in sign language recognition and synthesis, generative imaging, and human-computer interaction (HCI), resulting in papers and posters at ACCV Workshops, ICTAI, VISAPP, AICCSA, and UIST Adjunct.
2014 – 2016	Graduate Researcher, Graphics Jungle Lab University of Calgary Developed multiresolution methods for visualizing ICON climate data, resulting in publications at IEEE SciVis and VMV. Collaborating institutions: German Climate Computing Center (DKRZ) and National Center for Atmospheric Research (NCAR)

Teaching Experience

2025	Instructor, AI by Hand Workshop Series Led sessions on the mathematical foundations of recent AI research papers.
2024 – Present	Teaching Assistant, University of Colorado Boulder Course: Computer Vision (Fall 26, Spring 26, Fall 25, Spring 25, Fall 24) Average enrollment: 80–100 students
2017 – 2022	Assistant Professor, Ahsanullah University of Science and Technology Courses: Computer Graphics (4 terms) Image Processing (3 terms) Full-time teaching-focused faculty position; taught undergraduate students, designed course content and assessments, delivered weekly lectures, held office hours, managed the online learning platform, and completed course administration. Taught two to three sections per term; average enrollment: 80–130 students. Teaching evaluation: 4.83/5.00.
2014 – 2016	Teaching Assistant, University of Calgary Course: Python Programming (4 terms) Average enrollment: 50–60 students

Skills

Python, MATLAB, OpenGL, WebGL, JavaScript, TypeScript, C, C++

Research Software

2026	<i>byhand CLI</i> , Creator and Developer Developed a terminal-based tool for visualizing GGUF model architecture, including attention operations, tensor shapes, equations, and parameter sources. Project links: https://imruljubair.github.io/byhand/
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Selected Publications

<https://scholar.google.com/citations?user=H4-yZ3wAAAAJ&hl=en>

Preprints and Technical Reports

- 2026 Mohammad Imrul Jubair and Tom Yeh. The Architect: Interactive Visualization of Deep Learning Mathematics Directly in Microsoft Excel. Technical report, arXiv:2608.13572 [cs.HC].

Peer-Reviewed Journal Articles

- 2024 Danielle F. Pace, Hannah T. M. Contreras, Jennifer Romanowicz, Shruti Ghelani, Imon Rahaman, Yue Zhang, Patricia Gao, Mohammad Imrul Jubair, Tom Yeh, Polina Golland, Tal Geva, Sunil Ghelani, Andrew J. Powell, and Mehdi Hedjazi Moghari. HVSMR-2.0: A 3D cardiovascular MR dataset for whole-heart segmentation in congenital heart disease. *Scientific Data* 11, 721.

Peer-Reviewed Conference Papers

- 2022 Mohammad Imrul Jubair, Md. Masud Rana, Md. Amir Hamza, Mohsena Ashraf, Fahim Ahsan Khan, and Ahnaf Tahseen Prince. Altering facial expression based on textual emotion. In Proceedings of the 17th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP), Volume 4: **VISAPP**, pp. 917–924.
- 2022 Emdadul Haque, Md. Faraz Kabir Khan, Mohammad Imrul Jubair, Jarin Anjum, and Abrar Zahir Niloy. Book cover synthesis from the summary. In 2022 IEEE/ACS 19th International Conference on Computer Systems and Applications (**AICCSA**), pp. 1-6.
- 2022 Simanta Deb Turja, Mohammad Imrul Jubair, Md. Shafiur Rahman, Md. Hasib Al Zadid, Mohtasim Hossain Shovon, and Md. Faraz Kabir Khan. Shapes2Toon: Generating cartoon characters from simple geometric shapes. In 2022 IEEE/ACS 19th International Conference on Computer Systems and Applications (**AICCSA**), pp. 1-8.
- 2020 K. M. Arefeen Sultan, Mohammad Imrul Jubair, Md. Nahidul Islam, and Sayed Hossain Khan. toon2real: Translating cartoon images to realistic images. In 2020 IEEE 32nd International Conference on Tools with Artificial Intelligence (**ICTAI**), pp. 1175–1179.
- 2016 Mohammad Imrul Jubair, Usman Alim, Niklas Röber, John Clyne, and Ali Mahdavi-Amiri. Icosahedral maps for a multiresolution representation of Earth data. In Vision, Modeling & Visualization (**VMV**), pp. 161–168.

- 2013 Mohammad Imrul Jubair and Prianka Banik. A technique to detect books from library bookshelf image. In 2013 IEEE 9th International Conference on Computational Cybernetics (ICCC), pp. 359–363.

Peer-Reviewed Workshop Papers

- 2022 Mohammad Imrul Jubair, Arafat Ibne Yousuf, Tashfiq Ahmed, Hasanath Jamy, Foisal Reza, and Mohsena Ashraf. DIY Graphics Tab: A cost-effective alternative to graphics tablet for educators. **AAAI 2022 Workshop**.
- 2020 Oishee Bintey Hoque, Mohammad Imrul Jubair, Al-Farabi Akash, and Md. Saiful Islam. BdSL36: A dataset for Bangladeshi sign letters recognition. In Proceedings of the Asian Conference on Computer Vision (ACCV) Workshops, pp. 71–86.

Posters and Extended Abstracts

- 2022 Mohammad Imrul Jubair, Ali Ahnaf, Tashfiq Nahiyen Khan, Ullash Bhattacharjee, and Tanjila Joti. PerSign: Personalized Bangladeshi sign letters synthesis. In the Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (**UIST '22 Adjunct**), article 25, pp. 1–3.
- 2015 Mohammad Imrul Jubair, Usman Alim, Niklas Röber, John Clyne, Ali Mahdavi-Amiri, and Faramarz Samavati. Multiresolution visualization of digital earth data via hexagonal box-spline wavelets. IEEE Scientific Visualization Conference (**SciVis**), poster.

Mentoring Experience

- 2023 Mentor for STEM Research Experience, University of Colorado Boulder
Mentored high school students through hands-on research projects and research methods.
- 2019 – 2022 Undergraduate Thesis Advisor, Ahsanullah University of Science and Technology
Supervised student teams conducting experimental research in computer vision and human–computer interaction and advised them on preparing research papers.

Fellowships

University of Colorado Boulder

- 2026 McLaren Futures Family Fellowship
- 2023 Endowed Founder’s Fellowship
- 2022 Early Career Professional Development Fellowship

Islamic University of Technology

- 2008 OIC Undergraduate Student Scholarship

Awards

University of Colorado Boulder

2026	Outstanding TA Award
2023	Departmental Service Award

University of Calgary

2016	Travel Award
2014, 2015	Research Award
2014	International Recruitment Graduate Award
2014	International Student Differential Fee Reimbursement Award

Service

University of Colorado Boulder

2024	Team Lead, Computer Vision Group at Imagine AI Lab
2024	Graduate Peer Mentor
2024, 2026	Computer Science Peer Mentor for Incoming PhD students
2023, 2024	Volunteer, Computer Science PhD Open House
2023	Mentor for Prospective PhD Applicants
2023	General Co-Chair, Student Symposium

Ahsanullah University of Science and Technology

2020	Committee Member, International Conference on Computer & Information Technology (ICCIT)
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Islamic University of Technology

2011	Vice President, Computer Society (IUTCS)
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Invited Talks and Guest Lectures

2025	Invited Talk: Demystifying AI Blackbox Artificial Intelligence in Clinical Practice Workshop, Kettering General Hospital University Hospitals of Northamptonshire NHS Group, UK
2025	Guest Lecture: State-of-the-Art Algorithms for Large Language Models Computer Vision course, University of Colorado Boulder
2016	Invited Talk: Multiresolution Visualization of ICON Data German Climate Computing Center (DKRZ)

Reference

Dr. Tom Yeh
Associate Professor, Department of Computer Science
University of Colorado Boulder
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